Architectural Evaluation and Experimental Blueprint for Core-Space Multi-Adapter Composition

Deep Understanding of the Synapta Framework

The Synapta project constitutes a pre-registered empirical investigation into prompt-level multi-adapter Low-Rank Adaptation (LoRA) composition, optimized for memory-constrained edge hardware, specifically the Apple Silicon Unified Memory Architecture (UMA).[1] The fundamental engineering problem the system attempts to resolve is dynamic, multi-domain generalization during inference without incurring the computational overhead of full fine-tuning or the memory redundancy of deploying multiple monolithic large language models (LLMs). Operating under severe constraints—an Apple M3 processor with 8GB of unified memory—the architecture limits itself to 4-bit quantized base models (Qwen2.5-1.5B-Instruct and Mistral-7B-Instruct-v0.3) and prompt-level routing heuristics.[1]

A rigorous reverse-engineering of the framework reveals its core operational mechanics and mathematical formulations. The central design decision resides in the custom RoutedLoRALinear module, which replaces standard linear layers in the transformer architecture. For a given input activation x at layer l, the forward pass computes a base model output $z_l = W_{base} \cdot x$ and an adapter injection vector m_l. The injection vector is computed as a dynamically weighted sum of the active adapters: $m_l = \sum_{i \in \text{top-K}} w_i \cdot (\alpha/r) \cdot (x \cdot A_{i,l}) \cdot B_{i,l}$, where w_i represents the prompt-level routing weight for the i-th adapter, α is the LoRA scaling factor, and r is the adapter rank.[1] Because the Apple UMA allows the CPU and GPU to share a physical address space, the system loads all $N = 20$ adapter weight matrices into memory at startup, facilitating zero-copy access and circumventing the PCIe bus transfer bottlenecks inherent to discrete GPU architectures.[1]

To prevent catastrophic representation collapse when multiple adapters are injected simultaneously, the architecture implements two distinct clamping mechanisms. The Weight-Cap mode imposes a hard upper bound on the routing coefficients, defined algebraically as $w_i^* = \min(w_i, c)$, where c is a global hyperparameter (typically 0.5). Conversely, the Norm-Ratio mode imposes a per-layer activation gate based on the L_2 norm. It computes a scaling factor $\gamma_l = \min(1.0, c \cdot ||z_l||_2 / (||m_l||_2 + \epsilon))$ and attenuates the

composite adapter signal proportionally if its magnitude exceeds the base model's activation magnitude by the specified ratio.[1] Routing decisions are governed by a GatedRouter, which leverages a lightweight embedding-based similarity search (via all-MiniLM-L6-v2) augmented with a confidence gate. The gating rule dictates that a single adapter ($K = 1$) is utilized if the top domain probability $p_1 > \tau_{single}$ and the gap to the second candidate $(p_1 - p_2) > \text{gap_threshold}$; otherwise, the system defaults to $K = 2$ composition, combining the top two experts.[1]

Explicit and Hidden Assumptions

The architecture is predicated on several critical assumptions that dictate its operational bounds and theoretical validity.

The primary explicit assumption is the hypothesis of linear subspace composability. The framework posits that the linear combination of distinct, independently trained LoRA adapter outputs directly correlates with a coherent semantic blending of domain knowledge. It explicitly assumes that the knowledge embedded within the Medical Diagnosis adapter and the Mathematics adapter resides in non-interfering, additive parameter subspaces, allowing their simultaneous activation to resolve intersectional queries. A secondary explicit assumption is the necessity of magnitude constraint. The implementation of the Weight-Cap and Norm-Ratio clamping mechanisms explicitly assumes that the primary failure mode of multi-adapter composition is activation magnitude explosion—specifically, that unconstrained addition injects excessive energy into the residual stream, causing the model to diverge from its pre-trained manifold.[1]

Beneath the explicit architectural choices lie profound hidden assumptions. The most structurally limiting is the assumption of prompt-level homogeneity. By assigning a static routing weight w_i for the entirety of a generative sequence, the system inherently assumes that the necessity for specific domain knowledge is uniformly distributed across all tokens. It assumes a token predicting a mathematical operator requires the exact same degree of legal expert activation as a token predicting a statutory definition. Furthermore, the evaluation framework harbors a hidden assumption regarding metric sufficiency. By relying strictly on cosine similarity derived from a 22-million parameter sentence transformer (all-MiniLM-L6-v2) to evaluate performance on multi-domain expert queries [1], the framework assumes that lexical and distributional semantic overlap is a valid, high-fidelity proxy for logical reasoning accuracy, factual correctness, and multi-step deduction.

Brutal Critique

While the empirical rigor, the pre-registered methodology, and the honest reporting of negative results (such as the failures of Hypotheses 0, 2, and 5) represent commendable scientific practice [1], the underlying mathematical and architectural paradigms of the Synapta

framework exhibit severe theoretical deficits. A rigorous evaluation at the frontier of machine learning research reveals that the methodology relies on mathematically naive assumptions, reinvents suboptimal paradigms, and utilizes evaluation metrics that render the findings unpublishable at top-tier venues (e.g., NeurIPS, ICLR).

The Mathematical Fallacy of Linear Composability

The most glaring theoretical weakness is the naive assumption that independently trained LoRA adapters can be linearly added at inference time to yield coherent multi-domain reasoning. The empirical results demonstrate that $K = 2$ composition on single-domain queries degraded performance ($\Delta = -0.011$) and that the multi-domain gain (+0.017) was marginal, highly volatile, and entirely dependent on domain-pair orthogonality.[1] The architectural response—clamping activation magnitudes—is a superficial patch for a fundamental linear algebra problem.

Recent literature strictly contradicts the assumption of linear composability. Parameter-efficient fine-tuning (PEFT) updates, when trained independently, do not naturally align in orthogonal subspaces.[2] The NeurIPS 2025 paper "LoRA vs Full Fine-tuning: An Illusion of Equivalence" proves that LoRA training introduces high-ranking "intruder dimensions"—singular vectors that are structurally dissimilar to pre-trained weights and highly specific to the fine-tuning data.[3] When the RoutedLoRALinear module performs the linear addition $\sum w_i(xA_i)B_i$, it is superimposing these unaligned, non-orthogonal intruder dimensions. Because the basis vectors (the column space of B_i and the row space of A_i) of different adapters are not shared, their linear combination creates catastrophic directional interference, known as anisotropy.[4] The interference is structural and geometric; it cannot be resolved by scalar clamping. A top researcher would immediately reject the methodology because bounding the magnitude of an activation vector does not correct the geometric misalignment of the weight vectors. If two vectors represent conflicting semantic directions in the latent space, scaling their magnitude by 0.5 merely produces a smaller hallucination; it does not resolve the subspace conflict.[4]

The Algebraic Irrelevance of the Clamping Mechanisms

The study dedicates significant effort to comparing the Weight-Cap mode and the theoretically motivated Norm-Ratio mode.[1] However, as the empirical results in Phase 2b revealed, the Norm-Ratio clamp produced zero measurable benefit over the Weight-Cap ($\Delta_{SIM} = -0.0003$).[1] The report correctly notes that the clamp was "effectively inactive" because $Y_l \approx 1.0$ everywhere.

This result was entirely predictable from first principles. At rank-16, the L_2 norm of the

adapter activations ($||m_l||_2$) is mathematically bound to be infinitesimally small relative to the high-dimensional base model activations ($||z_l||_2$) in a 1.5B parameter model. The intrinsic dimensionality of the residual stream dwarfs the rank-16 projection. Designing a complex, dynamically calculated per-layer norm ratio that mathematically cannot trigger under the stated operating parameters demonstrates a disconnect between the theoretical intent and the algebraic reality of the network's forward pass.

Architectural Obsolescence: Prompt-Level Routing

Synapta utilizes prompt-level routing, applying a static w_i across the entire generation sequence.[1] This is a severe architectural regression. The research community established through frameworks like X-LoRA (2024) and MoE-LoRA that token-level routing is mandatory for complex reasoning tasks.[5] Multi-domain reasoning is not a homogeneous blend of facts; it is a sequential, combinatorial process. A query requiring mathematical proof and Python code generation does not require both domains simultaneously at every token. A token initiating a for loop requires the Python expert, while a token evaluating a derivative requires the Math expert. Prompt-level routing forces both adapters to remain active globally, guaranteeing continuous noise injection into domains where one adapter is irrelevant at that specific timestep. The confidence-gated embedding router [1], while functional for triage, is a rudimentary reinvention of early MoE gating networks and sentence-level sparse routing techniques (e.g., LoraHub, 2023) [7], lacking the granularity required for state-of-the-art inference.

Metric Inadequacy and Unpublishability

The framework's fatal flaw regarding publishability is its evaluation protocol. The study relies on semantic similarity using a 22-million parameter sentence transformer (all-MiniLM-L6-v2) to evaluate the outputs of a 1.5B and 7B parameter model.[1] This metric is fundamentally inadequate for evaluating advanced capabilities. Semantic similarity measures lexical distribution and bag-of-words overlap; it is entirely blind to logical soundness, factual accuracy, and mathematical correctness.

In the current LLM evaluation frontier (2025–2026), evaluating multi-domain logic requires rigorous, execution-based, or objective-graded benchmarks. Relying on an outdated embedding model to judge the multi-domain reasoning of modern LLMs renders the empirical findings suspect. An effect size of $+0.017$ in cosine similarity is indistinguishable from statistical noise in the embedding space and does not constitute a tangible improvement in actual intelligence density. To achieve publication at a top-tier venue, the evaluation must transition to objective, multi-step reasoning benchmarks (e.g., GPQA Diamond, SWE-Bench, ARC-AGI-2) that measure exact match or execution success rather than semantic proximity.[8]

Literature Frontier Analysis (2023–2026)

The transition from the heavily constrained Apple M3 UMA environment to a high-compute NVIDIA RTX 5090 (featuring 32GB of VRAM, PCIe Gen 5 bandwidth, and native FP8 Tensor Cores) necessitates a complete paradigm shift. The frontier of parameter-efficient multi-task learning, model merging, and adapter composition has evolved significantly from heuristic linear blending to mathematically grounded manifold optimization, core-space alignment, and inference-time reasoning architectures.

The Evolution of Parameter-Space Merging

The earliest attempts to solve cross-task adapter interference (2023–2024) operated directly in the weight space. Methods such as Task Arithmetic, TIES-Merging, and DARE hypothesized that interference was caused by parameter redundancy and sign conflicts.[10] TIES-Merging implemented a "TRIM, ELECT SIGN & MERGE" protocol, dropping low-magnitude parameters and resolving conflicting update directions via majority vote.[12] DARE extended this by applying random dropout to delta weights followed by magnitude rescaling.[10] AdaMerging advanced the paradigm by learning scaling coefficients directly from unlabeled test sets using unsupervised entropy minimization, adapting the merge dynamically.[13]

However, these weight-space methods suffer from a fundamental limitation: they operate in the full parameter space ($d \times d$) and frequently destroy the low-rank structure of the LoRA adapters.[15] When TIES-Merging sparsifies a reconstructed matrix $B \times A$, the resulting matrix is no longer rank-r, eliminating the computational efficiency that justified using LoRA initially. The problem of parameter interference was partially mitigated, but the underlying subspace misalignment remained unsolved.

The Subspace Alignment and Core-Space Paradigm (2025)

By 2025, the research consensus shifted toward enforcing orthogonal alignment during training or projecting updates into a shared mathematical basis before merging. This represents the current state-of-the-art for multi-adapter composition.

Method	Core Innovation	Resolution of Interference	Reference
LoRI (LoRA with Reduced Interference)	Freezes the projection matrix A as a shared random Gaussian	Forces all domain adapters into approximately orthogonal subspaces. Sparse	[16]

	initialization across all tasks.	masking on matrix B prevents parameter overlap.	
LoRA-LEGO	Disassembles LoRA into Minimal Semantic Units (MSUs) based on rank-wise parameter clustering.	Allows rank-wise assembly of multi-domain knowledge without full-space concatenation.	[4]
Core Space Merging	Decomposes task adapters into a shared low-rank basis via SVD, creating a compact $r \times r$ "Core Space".	Merging occurs entirely in the Tr -dimensional space. Mathematically guarantees no information loss while reducing complexity from $O(n^3)$ to $O(n^2)$.	[15]
StelLA	Employs a three-factor decomposition USV^T. Constrains U and V to the Stiefel manifold during training via Riemannian optimization.	Guarantees strict orthonormality of input and output subspaces, eliminating intruder dimensions.	[22]

StelLA and Core Space Merging effectively solved the linear additivity problem that Synapta failed to address. By aligning the bases (U and V) and restricting the domain-specific updates strictly to a scaling matrix (S), these frameworks ensure that cross-domain interference is mathematically neutralized.

Directional Anisotropy and Token-Level Dynamics (2026)

While subspace alignment solved the basis mismatch, it did not solve the routing granularity or the directional sensitivity imbalance.

1. **Addressing Anisotropy (TARA):** The 2026 paper "Preference-Aligned LoRA Merging" (CVPR) identified that even with sufficient subspace coverage, task losses exhibit unequal sensitivity to different LoRA directions (anisotropy). TARA mitigates this by aligning merging weights using a preference-weighted cross-entropy pseudo-loss, ensuring direction-wise reweighting that preserves task-relevant subspaces.[4]
2. **Token-Level Efficiency (CoMoL):** The limitation of prompt-level routing has been entirely bypassed by Core Space Mixture of LoRA (CoMoL, arxiv 2026). Traditional MoE-LoRA architectures compute full $d \times d$ transformations for each expert, resulting in severe memory bloat and latency. CoMoL solves this by performing the token-level routing *exclusively within the compact core space*.[24] Instead of aggregating high-dimensional expert outputs, CoMoL dynamically weights the small $r \times r$ core matrices based on the current token's hidden state, merging them into a single core matrix *before* the main transformation.[25] This maintains the FLOP profile of a standard LoRA forward pass while achieving the expressivity of token-level MoE.

Real and Unsolved Gaps

Despite the synthesis of core-space merging and token-level routing, a critical gap remains at the frontier: **Real-time, long-horizon reasoning synthesis over orthogonal manifolds.** Current token-level models (like CoMoL) route based purely on the immediate hidden state of a single token. They operate via System 1 pattern matching. However, complex multi-domain queries require System 2 deliberation—planning, searching, verifying, and backtracking across different domain subspaces. Furthermore, hardware utilization on single-GPU high-VRAM setups (like the RTX 5090) for dense token-level core-space MoE remains highly memory-bound, requiring specialized Triton or FlashAttention-3 kernels to maximize FP8 Tensor Core throughput without saturating the VRAM bandwidth.[27] The challenge is bridging the mathematical elegance of orthonormal core spaces with the agentic, test-time compute scaling required for advanced logical deduction.

Breakthrough Hypothesis Generation

To achieve a 3x–5x capability improvement and fully leverage the 32GB VRAM and massive FP8 compute capabilities of the RTX 5090, the architecture must completely abandon static prompt-level linear addition. The following hypotheses represent non-incremental, realistically testable pathways to breakthrough performance at a NeurIPS/ICLR level.

Hypothesis 1: Dynamic Riemannian Core-Space Routing (The CoMoL-StelLA Synthesis)

- **Core Idea:** Combine the orthonormal Stiefel manifold training paradigm of StelLA [22] with

the token-level core-space routing mechanics of CoMoL.[24] We pre-train N domain experts utilizing a three-factor decomposition ($US_iV^\top$), where the projection bases U and V are shared globally and constrained to be orthonormal. At inference time, a lightweight neural router projects the token's hidden state to dynamically mix only the $r \times r$ core scaling matrices (S_i) of the respective experts.

- **Why it should work (First-Principles):** The root cause of catastrophic interference in LoRA composition is basis misalignment, driven by the emergence of intruder dimensions.[3] By enforcing shared orthonormal bases via Riemannian optimization, all domain knowledge is structurally decoupled and forced into an aligned manifold. Routing strictly in the $r \times r$ core space requires negligible FLOPs, enabling dense, per-token expert blending without hitting the memory bandwidth bottlenecks that typically cripple MoE architectures.
- **Why existing methods fail:** Synapta linearly adds full-space, misaligned updates, causing destructive geometric interference. Standard MoE-LoRA computes full high-dimensional transformations for each expert, resulting in memory bloat and latency that scales linearly with the number of active experts K.
- **The 10x Factor:** This architecture enables $K = 8$ or $K = 16$ expert blending per token at the exact same computational cost as a single standard LoRA forward pass. It fundamentally solves the intelligence-density versus latency tradeoff, allowing massive domain scaling on a single RTX 5090.

Hypothesis 2: Speculative Subspace Tree Search for System 2 Reasoning

- **Core Idea:** Shift the multi-adapter composition problem from the weight space to the activation space by utilizing the massive FP8 throughput of the RTX 5090 to run parallel, speculative reasoning traces across different domain subspaces. Instead of mixing experts statically or at the token level, the base model acts as an agentic generator, spinning up a Monte Carlo Tree Search (MCTS) where different branches are conditioned entirely on different, isolated orthogonal expert adapters. A verification head scores the logical coherence of each branch, accepting the optimal multi-domain path.
- **Why it should work (First-Principles):** Multi-domain reasoning is rarely a homogeneous, simultaneous blend of concepts; it is sequential. Solving a bio-informatics problem requires first applying biological principles, then mathematical modeling, and finally algorithmic coding. Searching over discrete expert trajectories maps perfectly to test-time compute scaling, allowing the model to deliberate and backtrack.
- **Why existing methods fail:** Current composition methods (including TIES and AdaMerging) attempt to solve reasoning via parameter arithmetic.[11] Reasoning is a

sequential generation problem, not a weight-averaging problem. Averaging a legal adapter and a math adapter does not produce a model that can sequence legal logic followed by math logic; it produces a model that hallucinates heavily.
- **The 10x Factor:** By shifting from System 1 parameter mixing to System 2 test-time compute scaling (similar to the o1/o3 paradigm [28]), the model leverages the parallel processing power of the 5090 to verify its own logic, exponentially increasing performance on hard reasoning benchmarks.

Hypothesis 3: Latent Orthogonal Projection (LoRI-FP8) with Normalized Entropy Gating

- **Core Idea:** Implement the LoRI methodology [16] by freezing a single, randomly initialized projection matrix A for all tasks, and train the B_i matrices using extremely sparse masks (e.g., 90-95% sparsity). We replace Synapta's embedding router with an autonomous, mathematically derived Normalized Entropy Gate [1] that calculates the informational complexity of the prompt's hidden state. If the entropy exceeds a threshold, the system triggers a highly optimized, FlashAttention-3 powered FP8 kernel to compute the sparse updates across multiple orthogonal domains simultaneously.[27]
- **Why it should work (First-Principles):** If A is fixed and B_i is highly sparse, the parameter updates for different domains map to orthogonal vectors in a high-dimensional space. The dot product of their updates approaches zero, mathematically guaranteeing near-zero cross-task interference during merging, even without complex Stiefel manifold training.[16]
- **Why existing methods fail:** Synapta's unclamped and capped additions force unaligned, dense matrices to sum, destroying the low-rank structure and injecting noise.
- **The 10x Factor:** This achieves zero-shot adapter merging without any runtime neural routing overhead or layer-wise projection costs, maximizing batch size and continuous throughput on the RTX 5090 for extreme-scale multi-tenant serving.

Detailed Experiment Design: The CoMoL-StelLA Synthesis

Selected Hypothesis: Hypothesis 1 (Dynamic Riemannian Core-Space Routing) provides the most rigorous, mathematically sound, and hardware-optimal pathway for a fundamental breakthrough. It directly addresses the flaws of the Synapta architecture while maximizing the specific capabilities of the RTX 5090.

1. Model and Architecture Details

- **Base Model Selection:** Qwen2.5-14B-Instruct.
- **Justification:** The 14B parameter class represents the optimal frontier for single-GPU inference. When quantized to FP8 (native to the RTX 5090's Blackwell/Ada architecture),

the base weights occupy approximately 14GB of VRAM. This leaves 18GB of VRAM available for the KV cache (enabling long-context reasoning), the orthogonal adapter matrices, and high-throughput batching. Qwen2.5 exhibits state-of-the-art dense generalist reasoning capabilities, serving as a powerful substrate.[28]

- **Adapter Architecture:** We implement a three-factor decomposition $US_iV^\top$ corresponding to the StelLA methodology.[23]
  - $U \in \mathbb{R}^{d\times r}$ and $V \in \mathbb{R}^{d\times r}$ are shared globally across all domains. They are strictly constrained to the Stiefel manifold ($U^\top U = I$, $V^\top V = I$) to guarantee orthonormality.
  - $S_i \in \mathbb{R}^{r\times r}$ is the domain-specific core scaling matrix for domain i.
- **Routing Architecture:** A lightweight Multi-Layer Perceptron (MLP) router $R_\theta(x)$ is deployed at each transformer layer. It operates not on the massive d-dimensional hidden state, but on the projected low-rank input state $\hat{x} = V^\top x \in \mathbb{R}^r$.[25] It outputs a probability distribution $p \in \mathbb{R}^N$ over the N experts.

The Forward Pass Pipeline:

1. Projection: $\hat{x} = V^\top x$ (Computed once per token).
2. Routing: $p = \mathrm{Softmax}(R_\theta(\hat{x}))$
3. Core Merging: $S_{merged} = \sum_{i=1}^{N} p_i S_i$ (Dynamic $r \times r$ matrix addition).
4. Expansion: $m = U(S_{merged}\hat{x})$
5. Residual Addition: $output = z_{base} + m$

2. Data Strategy

Synapta's reliance on 40 synthetic questions is statistically void. We require robust, domain-specific, and multi-domain datasets.

- **Domain Expert Training:** To train the core matrices S_i, we utilize high-quality, domain-specific subsets from established 2025/2026 reasoning benchmarks: SciCode for scientific coding, the GPQA train split for physics/biology/chemistry, and Scale SEAL for legal/finance.[8]
- **Synthetic Routing Data (Multi-Domain):** We must train the router to switch domains mid-generation. We will generate 50,000 synthetic reasoning trajectories using a teacher model (e.g., DeepSeek-R1-Distill-Qwen-32B or Claude 3.7 Sonnet). We prompt the teacher to solve complex multi-domain queries and explicitly output specialized

<domain_x> tags before each discrete reasoning step. This creates supervised targets for the token-level router.

- **Data Scaling:** The training distribution must be uniformly balanced across 10 distinct domains. An imbalanced dataset will cause the router to collapse, perpetually favoring the majority class.

3. Training Plan and Pipeline

The training proceeds in three distinct, mathematically isolated phases to ensure stability.

- **Phase 1: Shared Basis Initialization (Stiefel Manifold)**
  - Initialize a single, global LoRA adapter on a uniform mixture of all domain data.
  - Utilize a Riemannian optimizer (converting standard Euclidean gradients via QR decomposition retraction) to learn the optimal orthonormal bases U and V.[22]
  - Upon convergence, completely freeze U and V.
- **Phase 2: Core Matrix Specialization**
  - For each domain i, initialize the $r \times r$ core matrix S_i randomly.
  - Train S_i exclusively on domain i's dataset. Standard Euclidean optimization (AdamW) is utilized here, as the core matrices are unconstrained.
- **Phase 3: Router Training (The Core Space)**
  - Freeze the base model, U, V, and all expert core matrices S_i.
  - Initialize the router R_θ. Train using the synthetic multi-domain dataset where the target expert is known per-token.
  - **Loss Function:** Standard causal language modeling cross-entropy loss, augmented with a load-balancing auxiliary loss to penalize the router if it collapses to a single expert: $L_{aux} = \beta N \sum_{i=1}^{N} f_i P_i$ (where f_i is the fraction of tokens routed to expert i, and P_i is the mean router probability).

Compute Utilization Strategy (Maximize RTX 5090): We leverage FlashAttention-3 to exploit the hardware support for FP8 low-precision on Hopper/Ada architecture GPUs.[27] To maximize Tensor Core throughput, we will write custom Triton kernels to fuse the $V^\top x$ projection and the router MLP into a single kernel execution. Because the core matrices S_i are exceedingly small (64×64), they can remain strictly within the L2 SRAM of the GPU, avoiding global memory fetches and completely alleviating the memory bandwidth bottleneck that plagues traditional MoE routing.

Hyperparameters:

- Adapter Rank $r = 64$ (to provide sufficient representational capacity in the core space).
- LoRA Scaling Factor $\alpha = 64$.
- Router Learning Rate: $1e - 3$ with cosine decay.
- Core Matrix Learning Rate: $3e - 4$.
- Batch Size: Maximized to saturate 30GB of VRAM (leaving 2GB safety buffer).

4. Evaluation Criteria

The evaluation protocol must shift from semantic similarity to objective reasoning execution.

Benchmark	**Justification**	**Target Metric**
GPQA Diamond	Tests high-level, Google-proof scientific reasoning. Resistant to data contamination.	Exact Match (EM) Accuracy.
SWE-Bench Verified	Tests rigorous, multi-step execution, agentic behavior, and coding reasoning.	Pass@1 (Code execution success).
ARC-AGI-2	Tests fluid intelligence and generalization to novel logic patterns entirely outside the training distribution.	Exact Match Accuracy.
GSM8K (Sanity)	Evaluates if the multi-domain routing degrades pure single-domain mathematical reasoning.	Exact Match Accuracy.

Detecting Real Improvement vs. Noise:

Improvement is verified if exact match accuracy increases on objective benchmarks. Noise injection is detected by measuring the perplexity (PPL) on standard single-domain validation sets. If the perplexity of the CoMoL-StelLA model increases compared to a single-domain expert, the token-level router is injecting cross-domain noise (anisotropy) into the generation stream.

Success Criteria

For the experiment to be deemed a scientific breakthrough worthy of NeurIPS/ICLR publication, rather than an incremental engineering patch, it must meet the following strict thresholds:

1. The Reasoning Gain Threshold

- **Metric:** Accuracy on the GPQA Diamond and SWE-Bench Verified benchmarks.
- **Threshold:** An absolute performance improvement of $\geq 15\%$ over the base Qwen2.5-14B zero-shot performance, OR a $\geq 30\%$ **relative reduction** in error rate compared to standard prompt-level LoRA composition (the Synapta baseline).
- **Why Meaningful:** GPQA Diamond is exceptionally difficult; PhD-level experts score approximately 65%, and top frontier models saturate around 90%.[8] An improvement of 15% absolute demonstrates that core-space routing effectively synthesizes domain expertise into actionable, multi-step deductive reasoning, allowing an open-weight 14B model to simulate the intelligence density of a much larger, proprietary parameter-class model.

2. The Zero-Interference Guarantee

- **Metric:** Accuracy on single-domain benchmarks (e.g., purely mathematical problems on GSM8K).
- **Threshold:** Degradation of $\leq 1.0\%$ compared to a dedicated, single-domain expert LoRA.
- **Why Meaningful:** This strictly proves that the Stiefel manifold training and core-space routing have successfully resolved the "intruder dimension" interference problem that mathematically plagued Synapta's $K = 2$ linear additions.

3. Failure Definition

The experiment is classified as a failure if it fails to beat the base model's zero-shot performance on multi-domain prompts by at least 10% absolute, if cross-task interference degrades single-domain performance by >5%, or if the token-level dynamic routing introduces a latency overhead exceeding 20% compared to a single standard LoRA forward pass.

Failure Analysis and Fallbacks

A rigorous, first-principles research methodology requires identifying precise failure vectors before initiating training and pre-registering technical fallbacks.

Failure Mode 1: Router Collapse

- **What could go wrong:** The token-level router R_θ may find a local minimum where it perpetually assigns $p = 1.0$ to a single "general knowledge" expert and 0.0 to all specialized experts, effectively ignoring the core matrices to minimize the auxiliary loss.
- **Early Detection:** Monitor the normalized entropy of the routing distribution p during Phase 3 training. If the average entropy drops below a critical threshold (e.g., $0.2 \log N$) early in the epochs, collapse is occurring.
- **Fallback Strategy:** Transition from soft-merging to a Top-K hard routing mechanism augmented with a noisy gating function (similar to classic MoE implementations). This forces gradients to flow through diverse experts, breaking the symmetry of the local minimum.

Failure Mode 2: Stiefel Manifold Capacity Bottleneck

- **What could go wrong:** Constraining the projection bases U and V to be strictly orthonormal across all highly disparate domains (e.g., Poetry vs. Python coding) might severely bottleneck the representational capacity of the adapter, preventing it from learning complex, domain-specific distributions.
- **Early Detection:** The Phase 2 training loss for the core matrices S_i plateaus prematurely at a high value, and single-domain benchmark performance falls significantly below standard unconstrained LoRA fine-tuning.
- **Fallback Strategy:** Relax the strict geometric constraint and pivot to the **LoRI methodology**.[16] Freeze V as a random Gaussian matrix (which provides approximate orthogonality in high-dimensional spaces without strict constraints) and allow unconstrained Euclidean updates to U and S_i.

Failure Mode 3: Hardware Utilization Inefficiencies

- **What could go wrong:** Despite theoretical efficiency, the custom Triton kernels responsible for the $r \times r$ core-space mixing might fail to effectively saturate the RTX 5090's Streaming Multiprocessors (SMs), resulting in a memory-bound bottleneck and high latency.

- **Early Detection:** Profiling the inference pass with Nsight Systems reveals low Tensor Core utilization (<30%) and high memory read/write latency during the dynamic routing step.
- **Fallback Strategy:** Revert from Token-Wise Routing to **Layer-Wise Routing** (following the X-LoRA paradigm).[6] Calculating the expert mixture once per transformer layer rather than once per token drastically reduces kernel invocation overhead and memory access frequency, while still providing significantly deeper representational flexibility than prompt-level routing.

Iteration Strategy

Scientific breakthroughs are rarely achieved on the first training run; they require pivoting intelligently based on empirical telemetry.

- **Scenario A: Success in Accuracy, Failure in Latency.**
  - *Observation:* The mathematical framework of Core Space Merging is validated (reasoning metrics improve), but the token-level dynamic calculation exceeds the 20% latency overhead bound.
  - *Pivot:* We must distill the routing probabilities. We will utilize the trained token-level router to label a massive dataset, and subsequently train a static, prompt-level predictor. This predictor will output a sequence of discrete expert states ahead of time (generating a "Routing Plan"). We can then inject the experts sequentially at the prompt generation phase using standard, highly optimized FlashAttention-3 kernels, separating the routing logic from the critical generation path.[27]
- **Scenario B: Perfect Domain Isolation, Failure in Reasoning Synthesis.**
  - *Observation:* The model successfully isolates domains (zero interference on single-domain tasks) and recalls isolated facts, but fails to logically link them on GPQA or SWE-Bench. The fundamental assumption that *adding independent domain parameter knowledge equates to reasoning* is proven false.
  - *Pivot:* Shift the paradigm entirely from *Parameter Merging* to *Activation Steering* (Representation Engineering). Apply "Conditional Activation Steering" techniques.[29] Instead of training matrices to recall domain facts, we will train task vectors that steer the model's intermediate hidden states toward specific logical operations (e.g., "deduction step," "verification step," "abstraction step").
- **Scenario C: Phase 1 Orthonormal Bases Fail to Capture Features.**
  - *Observation:* The shared Stiefel manifold training fails to converge on a globally useful basis for 20 disparate domains.
  - *Pivot:* Implement **TARA (Preference-Aligned LoRA Merging)**.[4] We abandon the pre-training geometric constraint. We train standard, independent LoRAs, but merge them post-hoc using the TARA preference-weighted cross-entropy pseudo-loss. This algorithm projects out the anisotropic "intruder dimensions" before inference, forcing alignment retrospectively rather than prescriptively.

By abandoning naive algebraic addition and fully embracing manifold-constrained core-space optimization and token-level dynamics, this experimental blueprint transitions the work from a heavily constrained edge-device engineering exercise to a profound theoretical advancement at the absolute frontier of machine learning reasoning architectures.

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